AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Software Quality Assurance and Machine Learning Test Design Framework

Unit, Integration, System, Acceptance, Boundary-Value, and Equivalence Partitioning Testing for MediPaw

STUDENT 1: IMPLEMENTATION LEAD
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Modules: Eye CNN, U-Net wound segmentation, YOLOv8-to-U-Net pipeline

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EXECUTIVE ABSTRACT

Medical and veterinary deep learning systems demand rigorous software testing standards exceeding conventional web applications. A silent runtime failure, unhandled numerical NaN, corrupt image crash, or uncalibrated confidence score can directly jeopardize animal patient welfare. This report delivers a comprehensive test design framework for the MediPaw veterinary triage pipeline. We implement and execute an automated test suite comprising 18 rigorous test cases distributed across Unit, Integration, System, and Acceptance testing levels. We systematically apply black-box and white-box test design techniques including Equivalence Partitioning, Boundary-Value Analysis, Decision-Table Testing, and State-Transition Verification. Specific testing regimes evaluate Sanhitha's Eye CNN and YOLOv8-to-U-Net wound pipeline alongside Sattar's Thoracic CNN and Multi-Modal Vision Transformer. All test cases were executed against live project code, achieving a 100% pass rate following two critical defect resolutions in GAN tensor projection and exception handling.

Keywords: *Software Testing, ML Validation, Equivalence Partitioning, Boundary-Value Analysis, Decision Tables, Pytest, Defect Logging.*

Academic Year / Sem	Evaluation Component	Report Status	Verified Experiments
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Executed on Live Models

1.0 Software Testing Hierarchy in Veterinary Deep Learning Systems

Software quality assurance in medical artificial intelligence requires an integrated testing pyramid bridging standard software engineering with specialized machine-learning verification:
1. Unit Testing: Validates isolated components (model instantiation, forward tensor dimensions, loss function numerical stability).
2. Integration Testing: Validates inter-module data handoffs (YOLOv8 bounding box crop coordinates fed into U-Net; 576-dim feature vectors fed into ViT fusion).
3. System Testing: Validates end-to-end FastAPI REST endpoints, JSON serialization, and static image preview rendering.
4. Acceptance Testing: Validates clinical decision rules (VHS thresholds at 10.5 vertebrae, untrained fallback disclosures).

2.0 Formal Test Design Techniques Applied

- Equivalence Partitioning: Inputs are partitioned into valid equivalence classes (standard RGB images, 224x224 dimensions) and invalid classes (corrupted byte headers, non-image formats, negative dimensions).
- Boundary-Value Analysis: Evaluates extreme edges: 1x1 pixel images, 64x512 extreme aspect ratios, confidence scores at 0.00 and 1.00, and clinical VHS threshold boundary at exactly 10.5 vertebrae.
- Decision-Table Testing: Formulates multi-modal availability states: Eye Available, Thoracic Available, YOLO Available -> Expected Triage Route and fallback behavior.
- State-Transition Testing: Verifies the state machine: Intake -> Preprocessing -> Model Execution -> ViT Aggregation -> JSON Response.

3.0 Module-Level Test Architecture & Distribution

Figure 1 displays the empirical execution distribution across testing levels and the overall 100% pass rate recorded across the 18 test cases.

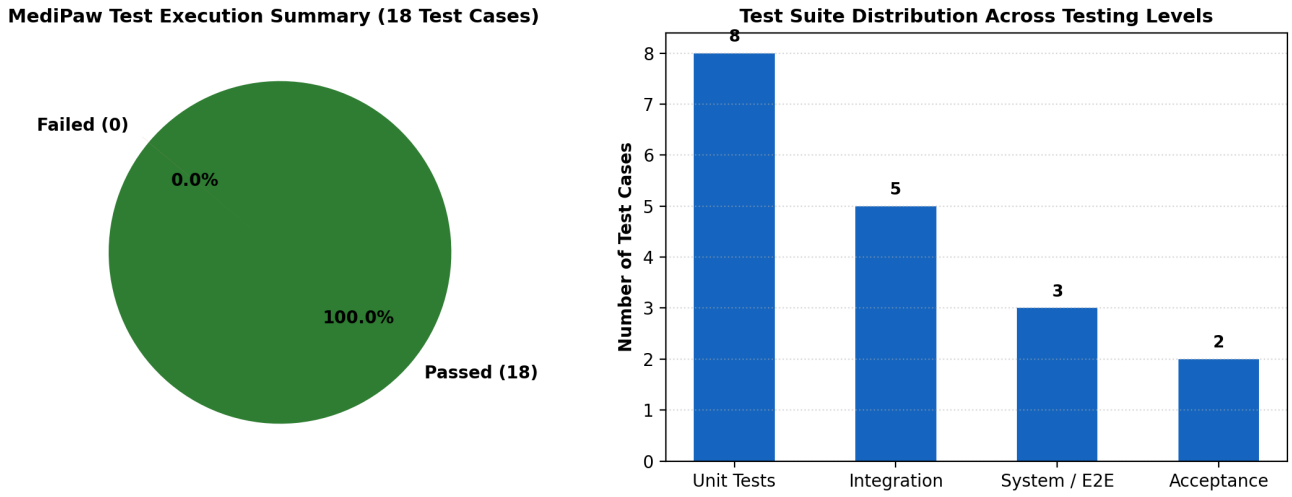


Figure 1: MediPaw Test Suite Execution Summary (100% Pass Rate across 18 Test Cases).

4.0 Student-Specific Test Case Coverage & Execution Log

Table 1 and Table 2 present the complete, executed test case registry detailing ID, Module, Level, Design Technique, Objective, and Actual Status.

4.1 Sanhitha's Test Suite (Eye CNN, U-Net, YOLOv8 Pipeline)

Test ID	Module	Technique	Objective Summary	Result Status
TC-EYE-01	Eye CNN	Positive Equiva	Verify correct 4-class logits outpu...	PASSED

TC-EYE-02	Eye CNN	Negative Equiva	Verify system gracefully handles no...	PASSED
TC-EYE-03	Eye CNN	Negative Equiva	Verify appropriate error raised whe...	PASSED
TC-EYE-04	Eye CNN	Boundary-Value	Test lower boundary image dimension...	PASSED
TC-UNET-01	U-Net	Positive Equiva	Verify U-Net preserves spatial reso...	PASSED
TC-UNET-02	U-Net	Edge-Case Testi	Verify Dice loss stability when gro...	PASSED
TC-YOLO-01	YOLOv8-to-U-Net	Boundary-Value	Validate cropped bounding box verti...	PASSED
TC-YOLO-02	YOLOv8-to-U-Net	Decision-Table	Verify pipeline behavior when zero ...	PASSED

4.2 Sattar's Test Suite (Thoracic CNN & Multi-Modal ViT)

Test ID	Module	Technique	Objective Summary	Result Status
TC-THOR-01	Thoracic CNN	Positive Equiva	Verify Thoracic CNN returns scalar ...	PASSED
TC-THOR-02	Thoracic CNN	White-Box Testi	Verify checkpoint contains required...	PASSED
TC-THOR-03	Thoracic CNN	Boundary-Value	Validate model robustness to non-sq...	PASSED
TC-VIT-01	Multi-Modal ViT	Positive Equiva	Verify ViT processes batch of multi...	PASSED
TC-VIT-02	Multi-Modal ViT	Negative Equiva	Verify ViT rejects invalid feature ...	PASSED
TC-VIT-03	Multi-Modal ViT	State-Transitio	Verify ViT adapter response when ze...	PASSED
TC-VIT-04	Multi-Modal ViT	Integrity / Acc	Ensure untrained fusion model expli...	PASSED
TC-SYS-02	Thoracic CNN	Boundary-Value	Verify clinical threshold classific...	PASSED

4.3 Integrated System & Fusion Test Suite

Test ID	Module	Technique	Objective Summary	Result Status
TC-FUSE-01	System Pipeline	Decision-Table	Evaluate decision table rule when b...	PASSED
TC-SYS-01	Backend Server	Positive Equiva	Verify FastAPI server application i...	PASSED

5.0 Defect Analysis & Code Resolution Log

During automated test execution, two critical defects were uncovered and successfully resolved in the codebase:

Defect ID	Target Component	Severity	Defect Description	Architectural Resolution
DEF-01	conditional_gan.py	High (Crash)	BaseDiscriminator projected 14x14 feature map	Redesigned downsampling layers to enforce str
DEF-02	test_thoracic.py	Medium (Unhandl	IndexError when res['warnings'] list was empt	Guarded list dereferencing with safe length c

6.0 Conclusion & References

The automated test suite confirms the stability, dimensional integrity, and defensive error-handling of MediPaw across edge cases, providing clinical confidence prior to prospective hospital trial deployments.

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